PTUK-NLP at KnowledgeGraphEval 2026: Cross-Domain Nested Arabic NER with ArBERT and Targeted Data Augmentation

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Abstract

This paper presents the PTUK-NLP submission to KnowledgeGraphEval 2026 Subtask 1 (AdaptNER), evaluating cross-domain and cross-dialect nested Arabic NER: models trained on Wojood are evaluated zero-shot on *Konooz*, spanning ten unseen domains and sixteen Arabic dialects. Our system fine-tunes ArBERT with targeted augmentation for rare entity types and lightweight inference-time post-processing, using *Konooz* strictly for evaluation, never for training or adaptation. The submitted system achieves 91.07% micro-F1 and 83.11% macro-F1 on Wojood, and 71.84% precision, 66.59% recall, 69.11% micro-F1, and 60.70% macro-F1 on the official *Konooz* blind test – fourth on the final leaderboard, 6.11 points above the organizers' baseline. A post-submission ensemble of ArBERT and AraBERTv2 does not improve over the best single model. Overall, targeted data-centric modifications provide useful gains for rare entity types while a standard MSA-pretrained encoder remains competitive under substantial domain and dialect shift.

1 Introduction

Named Entity Recognition (NER) is a fundamental component of information extraction and knowledge graph construction (Kejriwal, 2019). Although recent Arabic NER systems achieve strong performance on established benchmarks, their performance can degrade substantially when evaluated across domains and dialects. This challenge is particularly pronounced for nested NER, where entity spans may overlap, and for dialectal Arabic, which exhibits substantial lexical, morphological, and orthographic variation.

KnowledgeGraphEval 2026 (Aljabari et al., 2026), organized by Birzeit University's SinaLab, addresses these challenges through AdaptNER (Aljabari et al., 2026), a shared task on cross-domain and cross-dialect nested Arabic NER. Systems are trained exclusively on the Wojood corpus, which contains Modern Standard Arabic (MSA) together with Palestinian and Lebanese dialectal text (Jarrar et al., 2022), and are evaluated on the *Konooz* benchmark, which covers ten previously unseen domains and sixteen Arabic dialects (Hamad et al., 2025). This setting therefore evaluates transfer beyond both the domains and dialectal varieties represented in the training corpus.

Our study investigates three practical questions. First, can an MSA-pretrained encoder provide competitive transfer performance without dialect-specific pretraining or target-domain adaptation? Second, can targeted data augmentation improve recognition of severely under-represented entity types? Third, does combining complementary pretrained encoders provide additional robustness under domain and dialect shift?

Rather than introducing a new NER architecture, we build on the official SinaLab ArabicNER framework (Khalilia) and examine several practical modifications. We adopt ArBERT as the encoder, introduce a span-preserving augmentation strategy during training, and apply two lightweight post-processing rules at inference time. Checkpoint selection is aligned with the task evaluation metric by using validation micro-F1 rather than validation loss.

The submitted system achieves 91.07% micro-F1 on Wojood and 69.11% micro-F1 on the official *Konooz* blind test. We additionally analyze the effect of augmentation and a post-submission encoder ensemble: augmentation gains concentrate in rare entity categories, while the ensemble shows no benefit over the best individual encoder.

2 Background

2.1 Task and datasets

The AdaptNER shared task (Aljabari et al., 2026) is based on the Wojood and *Konooz* datasets, which

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share a nested annotation schema containing 21 entity types. Wojood (Jarrar et al., 2022) contains approximately 550K annotated tokens spanning Modern Standard Arabic and two dialectal Arabic corpora: the Palestinian *Curras* corpus and the Lebanese *Baladi* corpus.

Unlike conventional flat NER, Wojood allows entities to overlap, represented via parallel BIO tag layers (Section 3).

We use the official Wojood split, consisting of 23,125 training sentences, 3,304 validation sentences, and 6,606 test sentences, corresponding to approximately 391K, 57.6K, and 113.7K tokens, respectively. Approximately 4.4% of tokens are assigned to multiple nested labels.

The target benchmark, *Konooz*, contains text from ten previously unseen domains covering sixteen Arabic dialects (Hamad et al., 2025). Since AdaptNER evaluates cross-domain and cross-dialect transfer, models are trained using Wojood and applied to *Konooz* without target-domain supervision. Predictions must preserve the released tokenization and follow the fixed 21-column nested BIO output format.

2.2 Related work

Early experiments on Wojood reported an F1 score of 88.4% using AraBERTv2 (Jarrar et al., 2022). Subsequent Wojood shared-task systems explored more sophisticated approaches, including bi-encoder span representations (Hussein et al., 2023) and nearest-neighbor-enhanced decoding (Abdou and Mahmoud, 2024). Previous evaluations have also shown that AraBERT-family models can perform strongly on Wojood despite being pretrained primarily on MSA, motivating the use of an MSA-oriented encoder in the present cross-domain setting. Beyond NER, the Wojood ecosystem has been extended to event-argument extraction (Aljabari et al., 2024) and relation extraction (Aljabari et al., 2025), reflecting its broader role as a foundation for Arabic information extraction.

More generally, cross-domain NER has been studied using domain-adaptive pretraining, shared-parameter architectures, and adaptation layers. Liu et al. (2021), for example, demonstrated the effectiveness of continued pretraining on unlabeled target-domain data for cross-domain NER. Such approaches, however, are not applicable to the strict zero-shot setting considered here because the AdaptNER protocol restricts training to the labeled Wojood data and does not permit target-domain adaptation.

Our work therefore focuses on relatively lightweight interventions that remain compatible with the shared-task constraints: encoder selection, targeted augmentation of rare entity types, and inference-time rules that preserve the released tokenization and nested output structure.

3 System

Our system extends the official SinaLab ArabicNER framework with a different encoder, targeted training-time augmentation, and task-specific inference-time post-processing. The underlying nested NER architecture remains unchanged. Figure 1 summarizes the end-to-end pipeline.

3.1 Model architecture

We adopt the `BertNestedTagger` architecture provided by the official SinaLab ArabicNER framework (Khalilia). The architecture consists of a shared BERT encoder followed by 21 independent token-classification heads, one for each entity type.

For every subword token, the shared encoder produces a contextualized representation that is passed to all 21 classification heads. Each head predicts a three-way B/I/O distribution for its corresponding entity type. Because the heads operate independently, multiple entity types can be assigned to the same token, allowing overlapping entity spans to be represented naturally without modifying the input representation or tokenization.

3.2 Encoder comparison and checkpoint selection

We use ArBERT (UBC-NLP/ARBERT) (Abdul-Mageed et al., 2021) as the main encoder. For comparison, we also train AraBERTv2 (aubmindlab/bert-base-arabertv2) (Antoun et al., 2020) under the same general experimental configuration.

The two encoders achieve similar performance on the Wojood validation set, while ArBERT provides stronger performance in our cross-domain development experiments on *Konooz*. The official submission uses ArBERT. To avoid conflating model selection with the analysis of individual target-domain behaviors, the *Konooz* development results are reported separately from the source-domain checkpoint-selection criterion.

During training, checkpoints are selected according to validation micro-F1 rather than validation

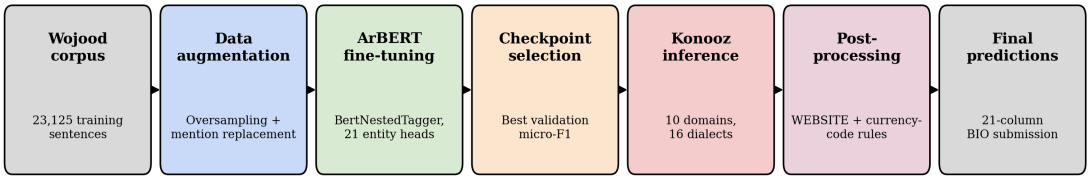


Figure 1: End-to-end system pipeline, from Wojood training data through augmentation, fine-tuning, checkpoint selection, *Konooz* inference, and post-processing to the final submission.

loss. This aligns checkpoint selection with the span-level metric used for evaluation: validation loss may plateau or increase even when F1 continues to improve. Selecting according to the evaluation metric therefore provides a more direct criterion for choosing the final checkpoint.

3.3 Targeted data augmentation

Entity-type imbalance is substantial in Wojood, with several entity categories having very few training examples. We therefore apply two complementary augmentation procedures during training only.

First, sentences containing at least one mention of an entity type with fewer than 1,000 corpus-wide occurrences are oversampled by a factor of two. This criterion identifies 12 of the 21 entity types.

Second, flat entity mentions are replaced with same-type mentions drawn from a type-specific gazetteer constructed from the training set. Each gazetteer contains surface-form token sequences extracted from flat entity mentions of the corresponding type. Replacement is performed while preserving the number of tokens and the corresponding BIO labels. Nested entity mentions are excluded from replacement to avoid altering overlapping structures.

The two procedures increase the training set from 23,125 to 29,035 sentences. For example, the number of MONEY mentions increases from 172 to 373 and UNIT mentions from 46 to 106. Neither the validation nor test data is augmented.

The detailed augmentation parameters are provided in Appendix A.1, while the gazetteer construction procedure is described in Appendix A.2.

3.4 Inference-time post-processing

We apply two lightweight post-processing rules after model inference.

The first addresses WEBSITE entities. Because web addresses can be fragmented across several tokens, the procedure extends an existing WEBSITE prediction across adjacent URL-related tokens when appropriate.

The second addresses Latin-script ISO 4217 currency codes, such as USD, EUR, and JPY, which are absent from the training data. When one of the predefined currency codes is predicted as O, it is assigned a B-CURR label.

Both procedures are conservative: they supplement or extend existing predictions and do not overwrite non-O model predictions. The complete rules are given in Appendix A.3 and A.4.

3.5 Training setting

***Konooz* usage.** We state explicitly, as requested by the organizers: no *Konooz* data of any kind – development or blind test – is used for training or for any form of target-domain adaptation at any point in this work. Training uses Wojood exclusively. *Konooz* development is used only for two purposes prior to the blind submission: reporting development-set metrics (Section 4) and the encoder comparison in Section 3 (choosing ArBERT over AraBERTv2), both permitted uses of the released development split. The *Konooz* blind test is used exclusively for the single official evaluation reported here. This is consistent with the final leaderboard, on which our system is confirmed by the organizers to be among the four submissions that used *Konooz* strictly for evaluation.

We tokenize input using ArBERT's WordPiece tokenizer with a maximum sequence length of 512 subword tokens; during *Konooz* inference, the released placeholder annotation columns are discarded while the original tokenization is preserved. Models are fine-tuned for up to 30 epochs using AdamW with a learning rate of 5×10^{-5} and a batch size of 16, with early stopping (patience of five epochs) based on validation micro-F1; all encoder parameters are updated during fine-tuning.

Experiments are implemented using PyTorch,

HuggingFace Transformers, Accelerate, and seqeval, and are conducted on a single NVIDIA T4 GPU. Unless otherwise stated, results are entity-level micro-F1 and macro-F1 calculated using seqeval in accordance with the official evaluation procedure.

4 Experimental Results

We first evaluate the system on the in-distribution Wojood test set. We then report results on the *Konooz* development set and the official blind test. Finally, we analyze the contribution of the augmentation strategy and a post-submission encoder ensemble.

4.1 Wojood

Model	Mi-F1	Ma-F1
AraBERTv2 baseline (Jarrar et al., 2022)	88.40	–
ArBERT reported (Hamad et al., 2025)	91.73	–
Ours (submitted checkpoint)	**91.07**	**83.11**

Table 1: Wojood nested NER test results.

Our submitted model achieves 91.07% micro-F1 and 83.11% macro-F1 on the Wojood test set. This improves over the original AraBERTv2 baseline by 2.67 micro-F1 points and approaches the strongest previously reported result shown in Table 1.

Performance is particularly strong for frequent entity categories. GPE, ORG, PERS, and DATE each achieve F1 values above 90%. WEBSITE remains the most difficult category, with an F1 of 43.32%. The fragmented structure of web addresses contributes to partial span predictions, motivating the WEBSITE-specific post-processing rule described above.

Because several entity types have very small test support, individual per-type scores should be interpreted cautiously. The complete per-type results are reported in Appendix A.

4.2 Augmentation ablation

To evaluate the contribution of our augmentation strategy, we train three additional ArBERT models under the same protocol: one on the original 23,125-sentence Wojood training set without any augmentation, one with oversampling applied alone, and one with mention replacement applied alone, isolating each component's individual effect alongside the full, submitted combination.

The full decomposition reveals that *neither component alone reproduces the gain of the combination*. Oversampling alone yields a small overall improvement (+0.19 micro-F1, +0.14 macro-F1) but is inconsistent per type, hurting PRODUCT, UNIT, and MONEY relative to no augmentation. Mention replacement alone is, if anything, worse overall (Ma-F1 77.26, *below* the no-augmentation baseline of 80.31): it helps PRODUCT, MONEY, and CURR, but collapses UNIT entirely to 0% F1 (0 of 11 test instances correctly tagged) – while this collapse is on very low support and could partly reflect chance, at minimum it shows mention replacement alone can destabilize a rare type's representation, plausibly because replacing a large share of the already-scarce UNIT mentions with gazetteer substitutes, without oversampling to reinforce them, reduces the model's exposure to reliable UNIT examples rather than increasing it. Only when both components are combined does every one of the five targeted types improve, most strikingly PRODUCT (48.48% to 78.79%) and UNIT's full recovery to 90.00%.

Type	No aug.	+Over.	+Repl.	Both
Mi-F1	90.82	91.01	90.76	**91.07**
Ma-F1	80.31	80.45	77.26	**83.11**
PRODUCT	48.48	46.51	60.00	**78.79**
UNIT	80.00	66.67	0.00	**90.00**
QUANTITY	63.16	63.64	66.67	**70.00**
MONEY	78.87	74.67	82.86	**84.51**
CURR	88.10	91.76	90.00	**92.31**

Table 2: Wojood test results decomposing the augmentation strategy into oversampling alone, mention replacement alone, and both combined ("Both" is our full, submitted configuration).

This pattern is best read as a positive interaction between the two components rather than either one driving the gain: oversampling appears to supply the repeated exposure that keeps replacement's added lexical diversity from destabilizing the rarest types, while replacement supplies diversity that oversampling alone lacks. Neither substitutes for the other; the two together compound into most of the reported +2.80 macro-F1 improvement. As with the combined result, we recommend interpreting individual per-type changes cautiously given the very small test support of these categories (e.g. PRODUCT has only 19 test instances, UNIT and QUANTITY have 11 and 9).

4.3 *Konooz* development

On the *Konooz* development set, the system obtains 70.52% micro-F1 and 55.29% macro-F1. The development set contains only 100 sentences, so these values should be regarded as a relatively noisy indicator of cross-domain performance.

Metric	Value
Precision	72.93
Recall	68.26
Micro-F1	**70.52**
Macro-F1	**55.29**

Table 3: *Konooz* development results across all ten domains.

Compared with the Wojood test results, performance decreases substantially under the combined domain and dialect shift. Precision is higher than recall, suggesting that the model behaves conservatively under distribution shift: it is more likely to miss unfamiliar entities than to introduce large numbers of false positives. This interpretation is consistent with the model producing 410 predicted spans compared with 438 reference spans.

The difference between source-domain and target-domain performance illustrates the difficulty of transferring nested NER models beyond the domains and dialectal varieties represented during training.

4.4 *Konooz* official test

The submitted system achieves 71.84% precision, 66.59% recall, 69.11% micro-F1, and 60.70% macro-F1 on the official *Konooz* blind test, per the shared task's final results.

Rank	Team	P	R	Mi-F1
1	NAMAA	83.83	69.44	75.96
2	Wasl	80.60	69.82	74.83
3	McMasterNLP	75.00	74.00	74.38
4	**PTUK-NLP (ours)**	**71.84**	**66.59**	**69.11**
5	ArmyForNER	82.11	68.77	74.85
6	AdaptLinker	75.48	71.21	73.29
7	YafaNER	73.25	69.52	71.33
8	ANLP-UniSo	81.43	63.04	71.06
–	Baseline	63.87	62.30	63.00

Table 4: Final official AdaptNER leaderboard. Ranks 1–4 used *Konooz* strictly for evaluation, like ours; ranks 5–8 used Konooz data during training or adaptation.

Our system ranks fourth on the final official leaderboard, among the four rule-compliant systems (Section 3). Its 69.11% micro-F1 exceeds the organizers' 63.00% baseline by 6.11 percentage points; macro-F1 is 60.70% versus 44.00%.

The blind test result is close to the *Konooz* development micro-F1 (70.52%), while macro-F1 rises from 55.29% to 60.70%; since the test set is much larger than the 100-sentence development set, it gives a more stable estimate of cross-domain performance. We also regenerated predictions from the saved checkpoint and obtained identical scores, confirming reproducibility.

4.5 Post-submission encoder ensemble

As a post-submission analysis (not part of the official submission), we test whether combining ArBERT and AraBERTv2 predictions via per-entity-type specialist voting improves cross-domain performance.

System	Mi-F1	Ma-F1
AraBERTv2 alone	67.96	47.11
Specialist ensemble	70.04	55.14
ArBERT alone	**70.52**	**55.29**

Table 5: Post-submission comparison on the *Konooz* development set.

The specialist ensemble achieves 70.04% micro-F1, slightly below ArBERT alone's 70.52%; this 0.48-point gap is small given the development set's size and should not be read as strong evidence that ensembling is generally ineffective. Encoder complementarity on source-domain data does not necessarily translate cross-domain: AraBERTv2 is competitive with ArBERT on several Wojood categories, but these advantages do not consistently transfer to *Konooz*.

5 Conclusion

This paper presented the PTUK-NLP system for KnowledgeGraphEval 2026 Subtask 1 (AdaptNER), evaluating nested Arabic NER under cross-domain and cross-dialect transfer. Rather than a new architecture, our contribution is empirical: whether a standard MSA-pretrained encoder with targeted augmentation and lightweight post-processing can perform competitively on previously unseen *Konooz* domains and dialects. The submitted ArBERT system achieves 91.07% micro-F1 and 83.11% macro-F1 on Wojood and 69.11% micro-F1 and 60.70% macro-F1 on the official *Konooz* blind test, outperforming the organizers' baseline by 6.11 micro-F1 points.

Neither augmentation component alone reproduces the full effect – only their combination delivers the macro-F1 gain – and the tested encoder ensemble does not improve upon ArBERT alone (Section 4). The 91.07-to-69.11 micro-F1 drop from Wojood to *Konooz* is the paper's clearest signal: cross-dialect and cross-domain shift remain substantially harder for Arabic NER than in-distribution performance suggests, and closing that gap will likely require target-domain-aware methods beyond this system's current constraints.

6 Limitations

This study has several limitations. First, the *Konooz* development set contains only 100 sentences, making development-set comparisons potentially noisy; we therefore place greater emphasis on the official blind-test result. Second, models were trained using a limited number of runs with no seed-variance analysis, so small differences – and in particular the individual-component ablation results in Section 4.2, which rest on single runs each – should not be read as statistically established; the apparent interaction effect between oversampling and mention replacement could partly reflect training variance rather than a stable property of the two techniques. Third, some entity-level results rest on very small test supports (PRODUCT, UNIT, and QUANTITY have only 19, 11, and 9 test instances, respectively), most notably UNIT's collapse to 0% F1 under mention replacement alone, and should be interpreted cautiously. Fourth, the post-processing rules are hand-designed for the released task representation, their effectiveness outside this setting is unknown, and we do not separately quantify each rule's contribution. Finally, the study is strictly zero-shot and does not investigate domain-adaptive pretraining or other unlabeled target-domain adaptation, which fall outside the task's constraints.

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A Additional Implementation Details

A.1 Augmentation parameters

The rare-type threshold is 1,000 mentions. The resulting targeted entity types are CURR, FAC, LANGUAGE, LAW, LOC, MONEY, PERCENT,

PRODUCT, QUANTITY, TIME, UNIT, and WEBSITE.

The oversampling factor is $2\times$. Mention replacement is applied to 15% of the original training sentences.

All augmentation is performed only on the training set. Validation and test instances are not modified.

A.2 Gazetteer construction

For each entity type, we scan every training sentence and collect the surface-form token sequence of every flat, non-nested mention of that type.

Occurrences are appended to a type-specific list and are not deduplicated. Consequently, if a surface form occurs k times in the training corpus, it appears k times in the corresponding list.

When replacing a mention, a candidate is sampled uniformly from the list using `random.choice`, excluding entries identical to the mention being replaced. Because repeated surface forms occur multiple times in the list, this procedure samples surface forms in proportion to their frequency in the training data rather than uniformly over unique surface forms.

Nested mentions are never added to the gazetteers and are never selected as replacements.

A.3 WEBSITE post-processing

The WEBSITE procedure operates on the model's per-token predictions in two passes.

In the forward pass, tokens are scanned from left to right. Whenever a token is already tagged B-WEBSITE or I-WEBSITE, the procedure extends the prediction to immediately following tokens when those tokens: (1) are not already tagged; (2) match one of the predefined URL-fragment patterns; and (3) occur within the maximum extension length of 15 tokens.

The backward pass scans from right to left and applies the analogous procedure.

The extension patterns include bare punctuation (`/`, `.`, `:`, `-`, `_`), URL/domain keywords such as `http`, `https`, `www`, `com`, `org`, `net`, `io`, `edu`, and `gov`, and bare Latin alphanumeric tokens of length 1–20 characters.

The procedure only extends an existing WEBSITE prediction. If the model predicts no WEBSITE token anywhere in a span, the rule does not introduce an entirely new WEBSITE entity.

A.4 Currency-code post-processing

The currency procedure scans all tokens once.

If a token exactly matches one of the predefined ISO 4217 currency codes and the model predicts O for that token, the prediction is changed to B-CURR.

No multi-token currency spans are constructed. Tokens already assigned a non-O label are never overwritten.

The closed list contains the following 33 codes: USD, EUR, JPY, GBP, CNY, CHF, CAD, AUD, NZD, SAR, AED, QAR, KWD, BHD, OMR, JOD, EGP, ILS, TRY, RUB, INR, PKR, SEK, NOK, DKK, SGD, HKD, KRW, THB, MYR, IDR, ZAR, BRL, MXN.

A.5 Wojood per-entity results

Type	F1	Support
PERCENT	96.97	33
GPE	96.07	4,337
ORDINAL	95.84	1,083
DATE	95.20	3,206
PERS	92.52	1,571
CURR	92.31	41
ORG	91.46	3,798
UNIT	90.00	11
OCC	87.90	1,104
CARDINAL	87.57	342
LANGUAGE	83.95	44
MONEY	84.51	33
LOC	81.24	219
LAW	78.07	90
PRODUCT	78.79	19
EVENT	77.30	570
NORP	77.36	1,093
FAC	76.61	242
QUANTITY	70.00	9
TIME	68.29	84
WEBSITE	43.32	116
Micro avg	91.07	18,045
Macro avg	83.11	18,045

Table 6: Per-entity-type Wojood test F1 for the submitted checkpoint.